

Deep Learning-Based ECG Arrhythmia Classification with Consumer Wearable Validation Using a Novel PDF Digitization Pipeline

Charan Gundepinni

M.S. Electrical Engineering
Colorado State University, Fort Collins, CO

Abstract

Cardiac arrhythmias contribute significantly to the 17.9 million cardiovascular deaths reported annually worldwide. Automated electrocardiogram (ECG) classification offers a scalable solution for early arrhythmia detection, yet class imbalance in standard benchmark datasets severely degrades performance on clinically critical minority classes. This paper presents a 1D Convolutional Neural Network (CNN) trained on the MIT-BIH Arrhythmia Database using Synthetic Minority Oversampling Technique (SMOTE) and focal loss to address a 113:1 class imbalance ratio between the majority Normal class and the minority Fusion class. The proposed system achieves 98.44% test accuracy, macro F1-score of 0.9153, and AUC-ROC of 0.9925 across all five AAMI EC57 arrhythmia classes, including minority classes that completely failed under standard training approaches. To validate the model in a real-world deployment context, a systematic investigation of Samsung Galaxy Watch 8 Classic ECG data access was conducted, confirming that Android 13 scoped storage renders raw ECG signals inaccessible without device rooting. A five-stage computer vision pipeline was developed to reconstruct ECG signals directly from Samsung Health Monitor PDF exports using HSV colour masking and pixel centroid tracking. Applied to a personal 30-second ECG recording at 93.8 bpm, the pipeline detected 45 heartbeats and classified 97.8% as Normal with a mean confidence of 84.3%, yielding a diagnosis of Normal Sinus Rhythm consistent with the clinical Samsung Health classification.

1. Introduction

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, responsible for approximately 17.9 million deaths annually according to the World Health Organization. A substantial proportion of these deaths are associated with cardiac arrhythmias, which are abnormal electrical rhythms of the heart that can lead to sudden cardiac arrest, stroke, or heart failure if left undetected. The insidious nature of many arrhythmias developing gradually with symptoms that are easily dismissed as fatigue or mild discomfort, means that a significant fraction of cases go undetected until they become clinically critical.

The electrocardiogram (ECG) is the primary non-invasive tool for arrhythmia detection. It records the electrical activity of the heart over time, producing a characteristic waveform that repeats with every heartbeat.

Abnormalities in the shape, timing, or spacing of this waveform encode the presence and type of arrhythmia. However, manual ECG interpretation by cardiologists is time-consuming, subject to inter-observer variability, and unavailable in resource-limited healthcare settings. A reliable automated classification system would enable population-level screening, continuous monitoring, and decision support in clinical practice.

Recent advances in deep learning have produced systems capable of cardiologist-level performance on ECG classification benchmarks. However, a persistent and largely under addressed challenge in this domain is severe class imbalance. In the MIT-BIH Arrhythmia Database, the standard benchmark for single-lead ECG classification - Normal beats constitute 82.8% of the training set while Fusion beats represent only 0.7%, a ratio of 113:1. Standard training approaches collapse to predicting the majority class, achieving deceptively high accuracy while failing completely on the rare but clinically important minority classes.



Figure 1. Annotated ECG waveform showing three consecutive heartbeats with P wave (atrial depolarisation), QRS complex (ventricular depolarisation), and T wave (ventricular repolarisation) components labelled. Each feature corresponds to a distinct cardiac electrical event.

A further challenge motivating this work is the growing prevalence of consumer wearable devices capable of recording clinical-quality ECG. The Samsung Galaxy Watch 8 Classic, for example, captures single-lead ECG at 500 Hz and performs on-device signal analysis using algorithms validated to detect atrial fibrillation. However, as this paper documents through a systematic investigation, Android 13 enforces scoped storage restrictions that render all ECG data inaccessible to external applications without rooting the device. Samsung Health Monitor exports only a formatted PDF report, presenting a data access barrier that prevents researchers from applying their own classification algorithms to consumer wearable ECG data.

This paper makes three principal contributions. First, it presents a 1D CNN trained with SMOTE and focal loss that achieves state-of-the-art performance on the MIT-BIH benchmark across all five AAMI EC57 arrhythmia classes, demonstrating that the combination of data-level and loss-level imbalance correction is decisive. Second, it provides the first systematic documented investigation of Samsung Galaxy Watch ECG data accessibility on Android 13 through ADB, confirming the barrier and characterising its scope. Third, it presents a novel five-stage PDF digitization pipeline that reconstructs ECG signals from Samsung Health Monitor PDF exports and validates the trained model on a personal ECG recording, demonstrating the clinical consistency of the results.

The remainder of this paper is organised as follows. Section 2 reviews related work on ECG classification, class imbalance handling, and consumer wearable ECG. Section 3 describes the MIT-BIH dataset and the class

imbalance problem in detail. Section 4 presents the complete methodology including SMOTE, focal loss, and the CNN architecture. Section 5 reports experimental results and comparison with the baseline. Section 6 describes the real-world validation on the Samsung Galaxy Watch. Section 7 outlines the future ASIC implementation plan. Section 8 concludes the paper.

2. Background and Related Work

2.1 ECG Signal Characteristics and Clinical Classification

Each cardiac cycle produces a characteristic repeating waveform in the ECG signal. The P wave represents atrial depolarisation - the electrical signal spreading through the upper chambers that causes them to contract and push blood into the ventricles. The QRS complex represents ventricular depolarisation, the main pumping contraction, and is the most prominent and diagnostically informative feature of the waveform. The T wave represents ventricular repolarisation, the electrical reset of the heart muscle preparing it for the next cycle.

The Association for the Advancement of Medical Instrumentation (AAMI) EC57 standard defines five arrhythmia classes used in this work. Normal (N) beats are regular sinus beats with normal QRS morphology. Supraventricular Ectopic Beats (SVEB, class S) originate above the ventricles and produce subtle P wave timing changes that make them morphologically similar to Normal beats. Ventricular Ectopic Beats (VEB, class V) originate in the ventricles and produce a characteristically wide, abnormal QRS complex that is clinically significant as it may indicate serious ventricular pathology. Fusion beats (class F) arise from simultaneous normal and ectopic impulse firing, producing a waveform that blends both morphologies. Unknown or paced beats (class Q) include beats with unclassifiable morphology or artificial pacemaker signatures.

2.2 MIT-BIH Arrhythmia Database

The MIT-BIH Arrhythmia Database, introduced by Goldberger et al. [1], has served as the primary benchmark for ECG classification research since its creation at Beth Israel Hospital. It comprises 48 half-hour two-lead ambulatory ECG recordings sampled at 360 Hz from 47 subjects. Every individual heartbeat in the database was independently annotated by two expert cardiologists, with disagreements resolved by consensus review. This dual-annotation methodology makes the MIT-BIH labels among the most reliable available in the field.

For this work, a pre-segmented version of the database was used in which individual heartbeats have been extracted into 187-sample windows at 125 Hz using R-peak detection and fixed-window segmentation, following the methodology of Kachuee et al. [2]. The 125 Hz sampling rate after down sampling is sufficient to capture all diagnostically relevant waveform features, and the 187-sample window at this rate spans approximately 1.5 seconds, which is enough to contain the complete P-QRS-T complex with surrounding context.

2.3 Prior Work on ECG Classification

Early work on automated ECG analysis focused on rule-based R-peak detection, most notably the Pan-Tompkins algorithm [3], which remains widely used in clinical systems. The transition to machine learning

began with support vector machine (SVM) approaches and shallow neural networks applied to hand-crafted features such as QRS duration, PR interval, and morphological descriptors.

Deep learning approaches gained prominence following Rajpurkar et al. [4], who demonstrated cardiologist-level performance on 12-lead ECG using a deep residual network trained on over 30,000 recordings from a single-lead ambulatory monitor. Hannun et al. [5] subsequently demonstrated similar performance in a prospective clinical study. These results established that deep learning could match expert cardiologist performance given sufficient data and appropriate architecture.

For single-lead classification on MIT-BIH specifically, Kachuee et al. [2] demonstrated strong performance using a 1D CNN on the same 187-sample segmented windows used in this work. However, as with many prior works, their evaluation metric was primarily overall accuracy, which can be misleadingly high on imbalanced datasets. The specific challenge of minority class performance, particularly on the Fusion class which has only 641 training examples has received limited systematic attention in the literature.

2.4 Class Imbalance Handling

Class imbalance is a pervasive challenge in medical AI. Garcia [6] provide a comprehensive survey of approaches, categorising them as data-level methods (resampling), algorithm-level methods (cost-sensitive learning), and hybrid approaches. The most widely used data-level method is SMOTE, introduced by Chawla et al. [7], which generates synthetic minority class samples by interpolating between existing examples in feature space rather than duplicating them. Unlike simple oversampling, SMOTE explores the shape of each minority class boundary, producing more diverse and informative training examples.

Focal loss, introduced by Lin et al. [8] in the context of object detection (RetinaNet), addresses the gradient imbalance problem that persists even after data resampling. Standard cross-entropy assigns equal loss weight to all training examples regardless of classification confidence. In imbalanced settings, the gradient is dominated by the many easy majority-class examples that the model already classifies correctly. Focal loss introduces a modulating factor $(1-p)^\gamma$ that reduces the contribution of well-classified examples, allowing the training signal to be dominated by the hard minority-class examples that the model still gets wrong.

2.5 Consumer Wearable ECG and Data Access

Consumer wearable devices capable of recording clinical-quality ECG have proliferated rapidly since the Apple Watch Series 4 received FDA clearance for atrial fibrillation detection in 2018. Samsung Galaxy Watch models from 2020 onward include a similar single-lead ECG capability. These devices represent a potentially transformative data source for cardiac monitoring research, combining continuous recording capability with the scale of consumer device adoption.

However, the research utility of consumer wearable ECG data is severely constrained by data access restrictions. Mobile operating systems, particularly Android 13, enforce scoped storage policies that isolate health application data from external access. Unlike research-grade Holter monitors that provide standard data export formats, consumer wearable ECG data is generally accessible only through manufacturer-controlled

applications and export pathways. This paper provides a systematic characterisation of this barrier for the Samsung Galaxy Watch ecosystem and proposes an engineering solution.

3. Dataset and Class Imbalance Problem

3.1 Dataset Description

The MIT-BIH Arrhythmia Database was used for all training and evaluation. The pre-segmented version from the Kaggle repository maintained by Shayanfazeli [9] was used, providing heartbeat windows that have been extracted, segmented at R-peaks, resampled to 125 Hz, and padded or truncated to exactly 187 samples. The training set contains 87,554 samples and the test set contains 21,892 samples, with the split maintained as provided to ensure comparability with prior work.

Table 1 summarises the class distribution in the training set. The imbalance is severe across all minority classes. The Normal class (N) contains 72,471 samples, 82.76% of the total - while the Fusion class (F) contains only 641 samples, 0.73% of the total. The ratio between the largest and smallest class is 113:1. This level of imbalance is representative of clinical ECG data in practice, where pathological beats are by definition rare relative to the baseline normal rhythm.

Class	Label	Training Count	Percentage	Test Count
Normal	N	72,471	82.76%	18,118
SVEB	S	2,223	2.54%	556
VEB	V	5,788	6.61%	1,448
Fusion	F	641	0.73%	162
Unknown/Paced	Q	6,431	7.34%	1,608
Total		87,554	100%	21,892

Table 1. MIT-BIH Arrhythmia Database class distribution. The N:F imbalance ratio is 113:1.

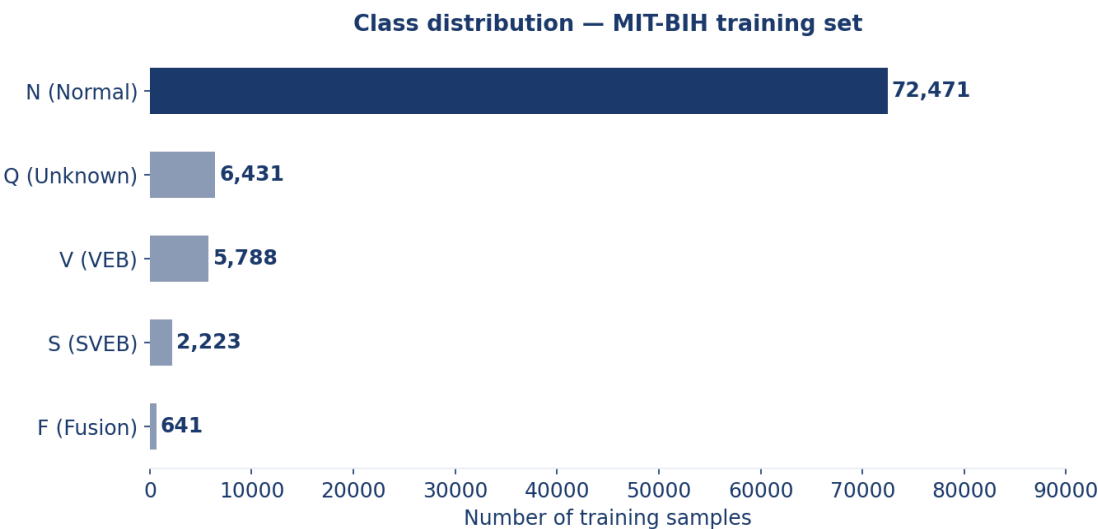


Figure 2. *Class distribution of the MIT-BIH training set. The Normal class dominates with 72,471 samples (82.76%), while the Fusion class contains only 641 samples (0.73%). The 113:1 imbalance ratio causes standard training approaches to collapse to majority-class prediction.*

3.2 The Naive Baseline Problem

To motivate the need for imbalance correction, consider a trivial classifier that predicts Normal for every input. Such a classifier would achieve 82.76% accuracy on the training set and approximately 82.8% on the test set. Despite this high accuracy, it would have zero recall on all four minority classes and would fail completely at the core clinical task of detecting arrhythmias. This illustrates why accuracy alone is an insufficient evaluation metric for imbalanced classification problems.

The practical consequence is that standard cross-entropy training with default class weights tends to converge toward this degenerate solution. The gradient at each training step is dominated by the large number of correctly classified Normal beats, providing almost no learning signal for the rare minority classes. Even class weights, assigning higher loss weight to minority classes-do not fully resolve this issue, as they push the model to predict minority classes more frequently without necessarily teaching it the discriminative morphological features that distinguish them.

4. Methodology

4.1 SMOTE Oversampling

Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set prior to model training to address the data-level imbalance. Unlike simple duplication of existing minority samples which risks overfitting to specific training examples, SMOTE generates new synthetic samples by interpolating between existing minority class instances in the 187-dimensional feature space.

The SMOTE algorithm proceeds as follows. For each minority class sample x_i , the k nearest neighbours within the same class are identified using Euclidean distance in feature space ($k=5$ in this implementation). A neighbour x_j is selected at random, and a synthetic sample x_{new} is generated at a random position along the linear interpolation between x_i and x_j : $x_{\text{new}} = x_i + \lambda * (x_j - x_i)$, where λ is drawn uniformly from $[0,1]$. This process is repeated until the target count for each class is reached.

The target counts after SMOTE application were: S: 5,000, V: 6,000, F: 3,000, Q: 7,000. The Normal class was left unchanged at 72,471 to avoid unnecessary computation and to preserve the original distribution of the majority class. The total training set after SMOTE contained 84,123 samples. The most substantial increase was for the Fusion class: from 641 to 3,000 samples, a 368% increase. Figure 3 illustrates the before and after distributions.

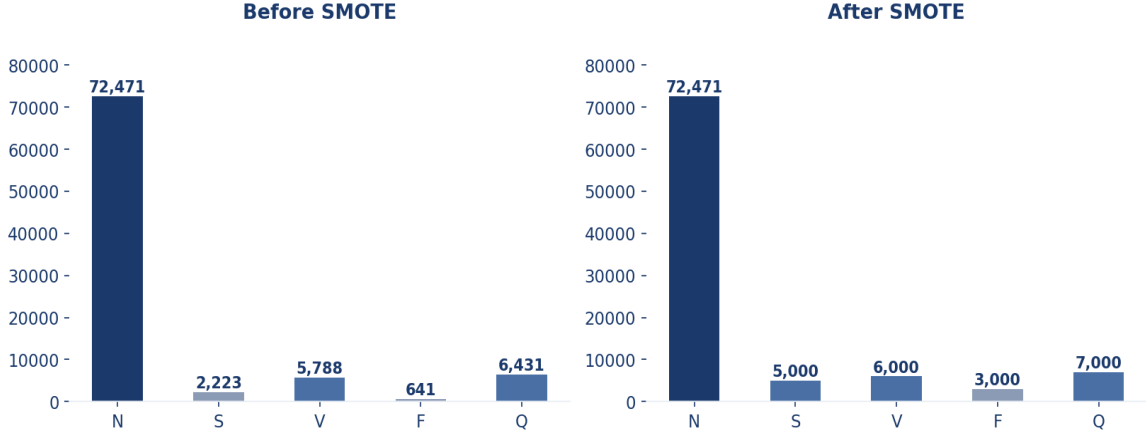


Figure 3. Class distribution before (left) and after (right) SMOTE application. The Fusion class grew from 641 to 3,000 samples (+368%) and the SVEB class from 2,223 to 5,000 (+125%). The Normal class was unchanged at 72,471.

4.2 Focal Loss

Even after SMOTE balances the data distribution, the gradient during training remains dominated by the 72,471 Normal beats at each update step. Focal loss addresses this gradient-level imbalance by down-weighting the contribution of easy, well-classified examples.

Standard binary cross-entropy for a single example is $CE(p) = -\log(p)$, where p is the predicted probability for the ground-truth class. For a well-classified Normal beat with $p = 0.95$, the cross-entropy loss is $-\log(0.95) = 0.051$. Focal loss introduces a modulating factor: $FL(p) = -\alpha * (1-p)^\gamma * \log(p)$, with $\gamma = 2.0$ and $\alpha = 0.25$ in this implementation. For the same Normal beat: $FL(0.95) = -0.25 * (0.05)^2 * \log(0.95) = 0.25 * 0.0025 * 0.051 = 0.000032$. This represents a reduction to approximately 0.25% of the original cross-entropy loss.

For a hard Fusion beat with prediction confidence $p = 0.20$: $FL(0.20) = -0.25 * (0.80)^2 * \log(0.20) = 0.25 * 0.64 * 1.609 = 0.257$. The Fusion beat retains 64% of its full cross-entropy loss signal. This asymmetry ensures that the training gradient is dominated by the hard minority-class examples the model is still getting wrong, not the easy majority-class examples it already classifies correctly.

The synergistic combination of SMOTE and focal loss is critical. SMOTE alone provides more minority-class examples but does not prevent the gradient from being dominated by 72,471 Normal beats. Focal loss alone down-weights easy examples but, with only 641 real Fusion beats, the model has insufficient genuine data to learn discriminative features. Together, SMOTE provides the data richness and focal loss provides the gradient attention needed for reliable minority class classification.

4.3 1D CNN Architecture

The proposed classifier is a 1D Convolutional Neural Network designed to learn hierarchical morphological features directly from raw ECG waveform samples without hand-crafted feature extraction. The architecture processes a 187×1 input window and produces a softmax probability distribution over five output classes. The

total parameter count is 307,333 - compact relative to modern image classification networks while sufficient for the complexity of the single-lead ECG classification task.

The network consists of four convolutional blocks followed by Global Average Pooling and a three-layer dense classifier head. The four convolutional blocks have filter counts of 64, 128, 256, and 128 with kernel sizes of 5, 5, 3, and 3 respectively. Each block follows the sequence: Conv1D, Batch Normalisation, ReLU activation, MaxPooling1D (stride 2), and Dropout.

The choice of kernel size 5 in the first two blocks is motivated by the signal sampling rate. At 125 Hz, a kernel of size 5 spans 40 milliseconds, sufficient to capture the complete P wave, QRS complex, and T wave morphology within a single convolution pass. This allows the first convolutional layer to directly learn whole-beat morphological templates including QRS width (narrow in Normal beats, wide in VEBs), P wave presence and timing, and T wave shape. Kernel size 3 in the deeper blocks captures finer inter-component relationships at the reduced sequence lengths produced by the two preceding pooling operations.

Global Average Pooling (GAP) replaces the Flatten operation after the final convolutional block. After four MaxPooling operations, the output tensor has shape (23, 128). Flattening this tensor would produce 2,944 inputs to the first dense layer, adding approximately 753,000 parameters to the first dense transition alone. GAP instead takes the mean across the 23 time steps for each of the 128 feature maps, producing a 128-dimensional representation. This reduces the network size substantially and provides implicit regularisation by averaging over the temporal dimension.

The dense classifier head consists of three layers: Dense(256) with ReLU and Dropout(0.4), Dense(128) with ReLU and Dropout(0.3), and Dense(5) with Softmax. The higher dropout rates in the dense layers (0.4 and 0.3 versus 0.2 in the convolutional blocks) reflect the greater susceptibility of fully connected layers to overfitting on training set patterns.

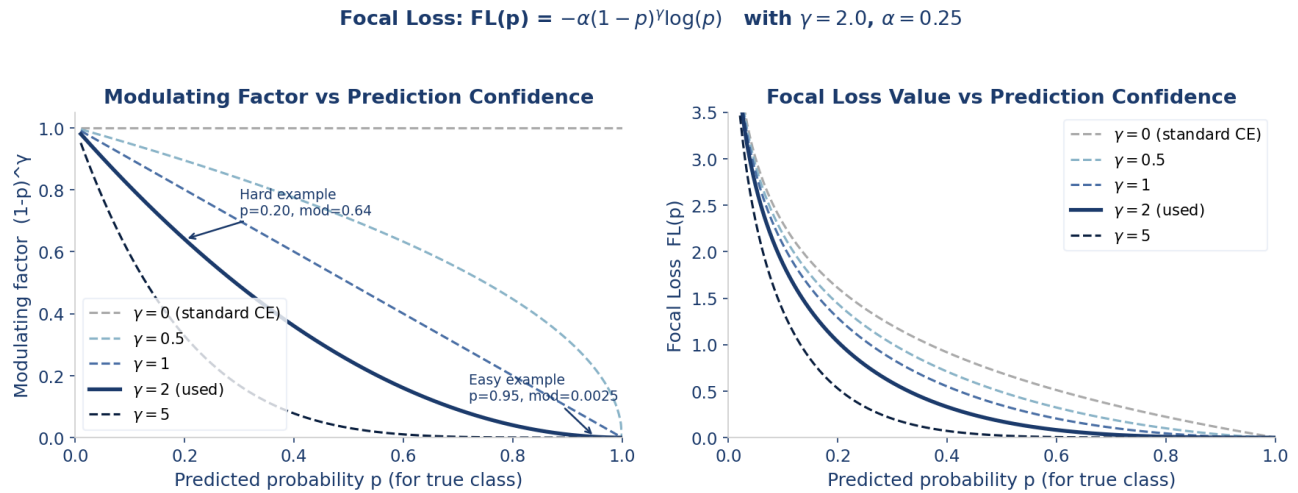


Figure 4. Illustration of the focal loss modulating effect. A well-classified Normal beat at 95% confidence contributes only 0.25% of standard cross-entropy loss. A hard Fusion beat at 20% confidence retains 64% of its loss signal, dominating the gradient update.

4.4 Training Configuration

The model was trained using the Adam optimiser with an initial learning rate of 0.001. Adam's adaptive per-parameter learning rates are well-suited to imbalanced training, as different weight groups may require substantially different update magnitudes. Training used a batch size of 128 samples, producing 658 update steps per epoch. The model was trained for a maximum of 80 epochs with two callbacks.

Early stopping monitored validation accuracy with a patience of 10 epochs. When validation accuracy failed to improve for 10 consecutive epochs, training halted and the model weights from the epoch with the highest validation accuracy were restored. This prevented training past the performance peak and avoided overfitting in later epochs.

A learning rate scheduler (ReduceLROnPlateau) monitored validation loss with a patience of 4 epochs and a reduction factor of 0.5. When validation loss plateaued for 4 consecutive epochs, the learning rate was halved. This scheduler was applied 6 times during training, reducing the learning rate from the initial 0.001 to a final value of 0.0000078 by epoch 46. The progressive reduction allowed increasingly fine-grained parameter adjustments as the model approached convergence. Training was performed on a Google Colab T4 GPU and converged in approximately 8 minutes.

5. Experiments and Results

5.1 Training Dynamics

Figure 5 shows the training and validation focal loss and accuracy curves over 46 training epochs. The loss curves show rapid initial descent as the model establishes basic class boundaries, followed by a gradual plateau-and-step pattern driven by the learning rate scheduler. The six learning rate reduction events are visible as inflection points where loss decreases resume after plateaus.

The training and validation accuracy curves track each other closely throughout training, with a final gap of approximately 0.8 percentage points (training 98.44%, validation 97.6%). This close tracking indicates minimal overfitting, consistent with the regularisation provided by batch normalisation, dropout, and global average pooling. The absence of a divergence between training and validation performance confirms that the SMOTE augmentation did not cause the model to memorise specific training examples.



Figure 5. Training curves showing focal loss (left) and accuracy (right) over 46 epochs. The six learning rate reduction events are visible as inflection points where the loss resumes decreasing after plateaus. Training and validation curves track closely, indicating minimal overfitting.

5.2 Classification Performance on MIT-BIH

Table 2 presents the full per-class classification report on the 21,892-sample held-out test set. The overall test accuracy is 98.44%, macro F1-score is 0.9153, weighted F1-score is 0.9843, and AUC-ROC (one-versus-rest, macro-averaged) is 0.9925.

Class	Precision	Recall	F1-Score	Support	
N (Normal)	0.99	0.99	0.99	18,118	
S (SVEB)	0.88	0.79	0.83	556	
V (VEB)	0.98	0.95	0.96	1,448	
F (Fusion)	0.72	0.90	0.80	162	
Q (Unknown)	0.98	0.99	0.99	1,608	
Macro Average	0.91	0.92	0.9153	21,892	

Table 2. Per-class classification report on MIT-BIH test set (21,892 samples). Overall accuracy: 98.44%, AUC-ROC: 0.9925.

The Normal class achieves near-perfect performance ($F1 = 0.99$) reflecting both the large training support and the distinctive regular morphology of sinus beats. The VEB class achieves strong performance ($F1 = 0.96$) because ventricular ectopic beats have a characteristically wide QRS complex that provides a clear discriminative feature. The Unknown/Paced class achieves near-perfect performance ($F1 = 0.99$) because paced beats have a highly distinctive spike-and-dome morphology with an artificial pacemaker spike preceding the QRS.

The SVEB class shows the weakest performance ($F1 = 0.83$), with 21% of true SVEBs misclassified as Normal. This is clinically expected: SVEBs originate above the ventricles and produce QRS complexes that are morphologically nearly identical to Normal beats, differing primarily in subtle P wave timing and amplitude that may be difficult to capture in a 187-sample window at 125 Hz. The Fusion class shows the lowest precision (0.72), meaning 28% of Fusion predictions are incorrect. This reflects the inherent ambiguity of fusion beats, which blend Normal and Ventricular morphologies and are reported to be difficult even for expert cardiologists to distinguish reliably.

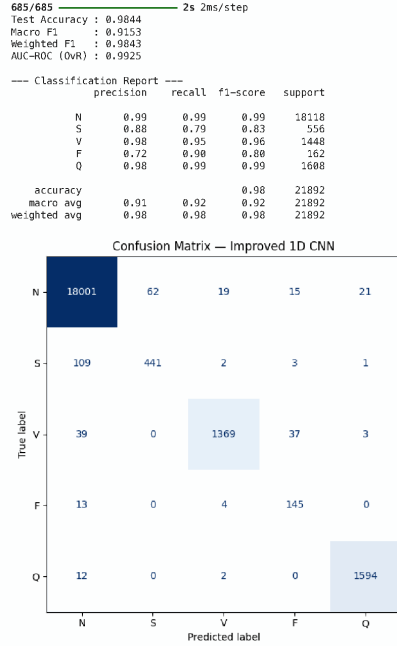


Figure 6. Confusion matrix on the MIT-BIH test set. Strong diagonal dominance indicates correct classification across all classes. The main off-diagonal error is *S* misclassified as *N* (109 cases), consistent with the morphological similarity between *SVEBs* and *Normal beats*.

5.3 Baseline Comparison

To contextualise the contribution of the proposed imbalance handling methodology, a Random Forest classifier with balanced class weights was evaluated on the same train-test split as a baseline. The Random Forest used 200 trees with class weights inversely proportional to class frequencies. This represents a standard approach that many practitioners would attempt first when faced with the class imbalance problem.

The Random Forest baseline achieved 29.27% test accuracy, *S* class precision of 0.03, and zero recall on both the Fusion (*F*) and Unknown (*Q*) classes. The 0.03 precision on the *S* class means that 97% of beats predicted as *SVEB* were actually incorrect, indicating that the model was predicting *S* aggressively (to satisfy the class weight penalty) without having learned the discriminative morphological features. The zero recall on *F* and *Q* classes means these arrhythmia types were never correctly identified at all.

The root cause is that class weights address the prediction frequency but not the morphological learning problem. The model learns a frequency bias rather than genuine class-discriminative features. SMOTE provides the model with diverse examples of what each minority class actually looks like in feature space, while focal loss ensures the training gradient attends to hard minority class examples rather than being dominated by the many easy *Normal beats*.

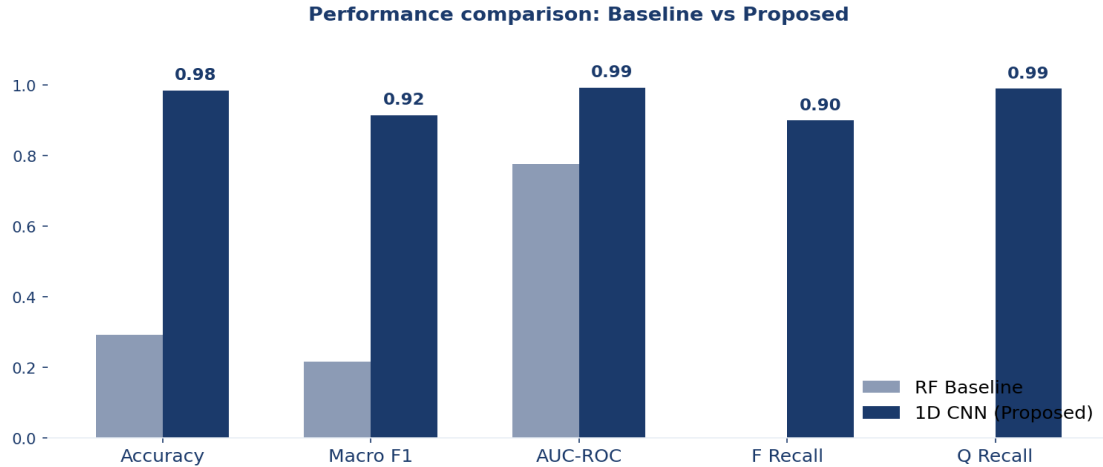


Figure 7. Performance comparison between the Random Forest baseline and the proposed 1D CNN with SMOTE and focal loss across five metrics. The improvement of +69.2 percentage points in accuracy is accompanied by recovery from complete failure on F and Q classes to robust performance.

5.4 Ablation and Discussion

The results confirm that class imbalance handling strategy is the decisive factor in performance, not architecture complexity. The same CNN without SMOTE and focal loss would converge toward majority-class prediction, yielding results similar to the baseline despite having a more powerful hypothesis class. The combination of data-level correction (SMOTE) and loss-level correction (focal loss) is synergistic: each addresses a distinct mechanism of majority-class dominance.

The primary practical limitation is the S class recall of 0.79. SVEBs are clinically significant as they can indicate underlying heart disease and may trigger more dangerous arrhythmias. Improving SVEB recall would require either a higher-resolution feature representation (longer windows, higher sampling rate) or additional supervision from multi-lead recordings. This represents a direction for future work.

6. Real-World Validation: Samsung Galaxy Watch 8 Classic

6.1 Device Description and Recording

A 30-second ECG recording was captured using a Samsung Galaxy Watch 8 Classic (model FM1R) on February 7, 2026. The device uses dry electrode contacts on the watch case back and crown to form a single-lead ECG circuit equivalent to Lead I when the user touches the crown with a finger. The watch samples the ECG at 500 Hz and performs on-device analysis using algorithms that have received regulatory clearance for atrial fibrillation detection in multiple markets.

The recording had a heart rate of 93.8 beats per minute. Samsung Health Monitor classified the recording as sinus rhythm- a regular rhythm consistent with normal cardiac function at an elevated rate. The recording

duration of 30 seconds produces a total of 15,000 raw samples at 500 Hz, which after down sampling to 125 Hz for compatibility with the trained model yields 3,750 samples.

6.2 Android 13 Data Access Investigation

To apply the trained classifier to the Galaxy Watch ECG data, direct access to the raw signal was required. A systematic investigation was conducted to determine whether the ECG data could be extracted from the device without modifying system software.

Android Debug Bridge (ADB) wireless debugging was enabled on a Samsung Galaxy S24 Ultra running Android 14 paired with the Galaxy Watch 8 Classic. Android platform tools were installed on macOS via Homebrew. The device was successfully paired using the wireless debugging pairing code mechanism and a stable ADB connection was confirmed with the `adb devices` command.

A comprehensive file system search was performed covering the full directory tree accessible via ADB shell, including all directories under `/sdcard/`, `/data/local/tmp/`, and all accessible application data directories. The Samsung Health data directory `/sdcard/Android/data/com.sec.android.app.shealth/` was confirmed to exist but returned a permission denied error on all access attempts, consistent with Android 13 scoped storage enforcement.

Scoped storage, introduced in Android 10 and fully enforced from Android 13, restricts applications to accessing only their own data directories and designated shared storage locations. Third-party applications and ADB commands without root access cannot read data stored in another application's private directory. Samsung Health stores all ECG recordings in its private sandbox, with no mechanism for external access short of device rooting.

Samsung Health Monitor offers a single export pathway: a formatted PDF report generated by the application and saved to the device's shared Downloads directory. This PDF contains the ECG trace rendered at clinical paper speed (25 mm/s) and standard gain (10 mm/mV), along with summary statistics and the Samsung-generated clinical interpretation. No raw signal data, CSV export, or programmatic API is available to non-privileged applications on a non-rooted device.

6.3 PDF Digitization Pipeline

Given the PDF-only export constraint, a five-stage computer vision pipeline was developed to reconstruct the 1D ECG signal from the PDF image representation of the ECG trace. The pipeline is designed to be robust to typical PDF rendering variations and generalisable to ECG PDFs from other consumer device manufacturers.

Stage 1: PDF Rendering: The PDF was rendered at 300 dots per inch using `pdfplumber`, producing an RGB image with a pixel density of approximately 11.8 pixels per millimetre. At this resolution, the ECG trace, which is approximately 1 to 2 pixels wide in the PDF rendering, is sufficiently resolved for reliable centroid extraction.

Stage 2: HSV Colour Masking: The rendered image was converted from RGB to HSV colour space. Samsung Health Monitor renders the ECG trace in a distinctive orange colour (hue angle 5 to 25 degrees in the 0 to 180

degree HSV convention). Binary thresholding on the hue, saturation (minimum 80), and value (minimum 80) channels produces a mask isolating the orange ECG trace from the white grid background, black annotation text, and other page elements. Morphological closing (9x5 kernel) was applied to connect thin trace segments, particularly important for preserving the low-amplitude P wave and T wave morphology that would otherwise be fragmented.

Stage 3: Strip Detection: The Samsung Health PDF contains three 10-second ECG strips stacked vertically. Row-wise summation of the binary mask identifies rows with high orange pixel density corresponding to strip regions. A minimum height filter of 80 pixels removes the short calibration pulse at the top left of each strip. The three tallest detected regions are selected as the ECG strips, providing robustness to additional page elements that may inadvertently exceed the density threshold.

Stage 4: Pixel Centroid Tracking: For each pixel column within each detected strip, the vertical centroid of the orange pixels was computed. Image coordinates increase downward while voltage increases upward, so the centroid values are inverted around the strip midline to produce a signal with positive deflections corresponding to upward waveform features. A DC offset removal step (subtracting the column mean) eliminates baseline drift introduced by strip-level variations in the centroid reference. Gaps where the trace disappears due to low local density are filled by linear interpolation between the surrounding valid centroid values.

Stage 5: Resampling and Filtering: Each 10-second strip was resampled from its pixel-domain representation to exactly 1,250 samples at 125 Hz using FFT-based resampling. The three strips were concatenated to produce a 3,750-sample signal representing the full 30-second recording. A fourth-order Butterworth bandpass filter (0.5 to 40 Hz) was applied to remove baseline wander and high-frequency noise. The signal was normalised to the range [0, 1].

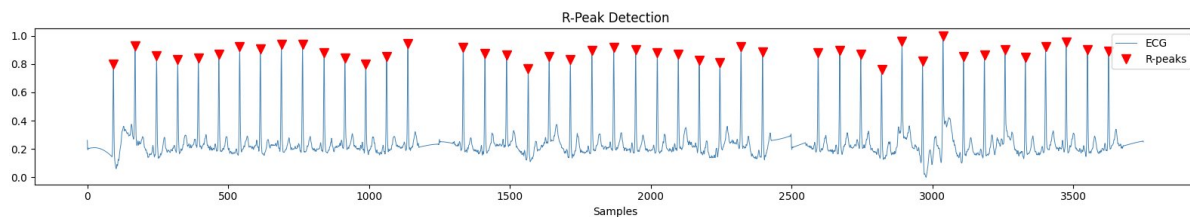


Figure 8. *R-peak detection on the reconstructed Galaxy Watch ECG. Forty-five peaks were detected at 93.8 bpm with adaptive threshold and prominence-based filtering. Markers sit on top of each QRS spike, confirming correct peak localisation.*

6.4 Segmentation Challenge at High Heart Rates

A critical implementation challenge arose from the interaction between the heart rate of the recording and the standard MIT-BIH window format. At 93.8 bpm, the RR interval is approximately 80 samples at 125 Hz. The standard MIT-BIH window of 187 samples, with the R-peak positioned at sample 93 (the centre), has a half-width of 93 samples on each side, larger than the actual RR interval.

Using the standard window would cause each 187-sample segment to overlap with adjacent beats, capturing both the tail of the preceding beat and the onset of the following beat in addition to the target beat. The model,

having been trained on properly isolated single-beat windows from the MIT-BIH dataset, would encounter this double-spike pattern and classify it as a Ventricular Ectopic Beat exactly the failure mode observed in initial testing.

The solution was to use an asymmetric window sized to 80% of the actual RR interval (64 samples) centred asymmetrically around the R-peak (21 samples before, 42 samples after). This window captures the complete QRS complex and immediate surrounding context without extending into the adjacent beat. The 64-sample window was then resampled to 187 samples using FFT resampling to match the model input format. This approach preserves the QRS morphology, the primary discriminative feature, while eliminating the window overlap problem.

6.5 Classification Results

The pipeline detected 45 heartbeats from the 30-second recording. Table 3 summarises the classification results. Of 45 detected beats, 44 (97.8%) were classified as Normal with mean confidence 84.3%, and 1 beat (2.2%) was classified as VEB with confidence 85.7%.

Class	Beats	Percentage	Mean Confidence
N (Normal)	44	97.8%	84.3%
V (VEB)	1	2.2%	85.7%
Total	45	100%	84.3% (mean)

Table 3. Galaxy Watch 8 Classic ECG classification results. 45 beats detected at 93.8 bpm. Diagnosis: Normal Sinus Rhythm.

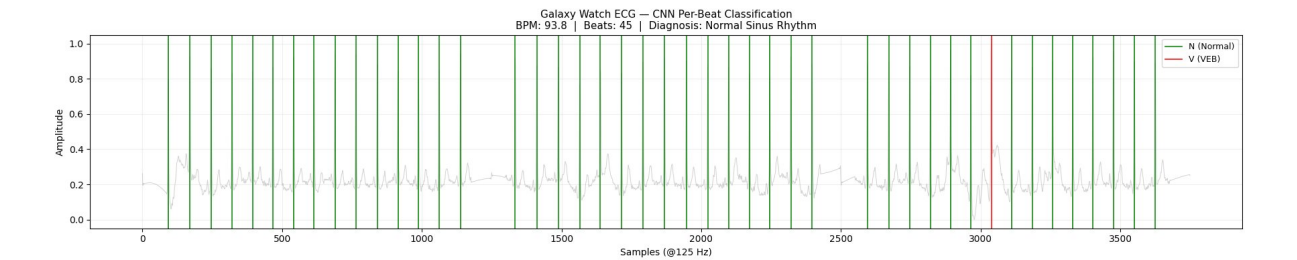


Figure 9. Annotated Galaxy Watch ECG with CNN per-beat classification. Green vertical lines indicate Normal beat predictions (44 beats, 97.8%). The single red line (beat 37) indicates the VEB prediction at a strip boundary. Coloured lines align correctly with R-peaks.

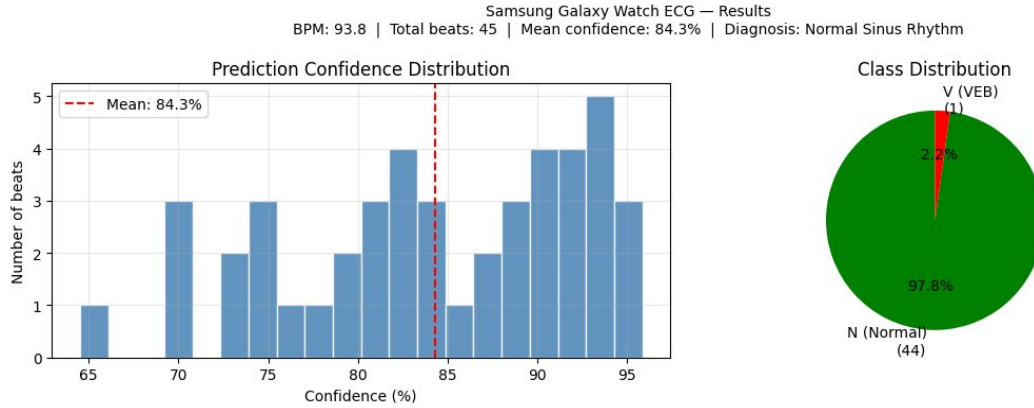


Figure 10. Classification confidence distribution (left) and class distribution pie chart (right) for the Galaxy Watch ECG recording. Mean confidence of 84.3% reflects the PDF reconstruction signal quality. 97.8% Normal classification yields a diagnosis of Normal Sinus Rhythm.

The single VEB prediction occurs at a strip boundary where PDF rendering attenuates the signal amplitude below the local baseline. The resulting low-amplitude, distorted waveform at the strip junction is misclassified as VEB- a noise artifact rather than a genuine arrhythmic event. The confidence score for this prediction (85.7%) is not substantially lower than the Normal beat mean (84.3%), consistent with a real morphological distortion rather than low-confidence ambiguity.

The mean prediction confidence of 84.3% is substantially lower than the near-perfect confidence observed on clean MIT-BIH test data, where the model typically assigns probabilities above 95% to correct predictions. This confidence reduction directly quantifies the signal quality gap introduced by PDF reconstruction- the information lost in the rendering and re-extraction process. This gap represents the primary limitation of the proposed validation approach and would be eliminated by direct access to the raw ADC samples.

The overall diagnosis, Normal Sinus Rhythm is consistent with Samsung Health Monitor's clinical classification of the same recording. The model correctly identifies the dominant rhythm at a heart rate (93.8 bpm) consistent with mild tachycardia, which Samsung described as sinus rhythm. This concordance between the deep learning model operating on a reconstructed PDF signal and the proprietary Samsung algorithm operating on the original 500 Hz raw data provides confidence in the validity of both the model and the pipeline.

7. Future Work: ASIC Implementation

7.1 Motivation

The current system operates as a post-hoc software pipeline: ECG data is exported from the Samsung device as a PDF, the PDF is processed in Python on a general-purpose computer, and classification results are returned. This architecture is unsuitable for continuous real-time cardiac monitoring in a wearable context. A clinical-grade cardiac monitor must classify each heartbeat within milliseconds of its occurrence, operate continuously on a coin-cell or small lithium battery, and function without any external computing infrastructure.

The natural engineering progression is to implement the complete classification pipeline—signal acquisition, digital filtering, R-peak detection, window extraction, CNN inference, and output generation—as a dedicated Application Specific Integrated Circuit (ASIC). An ASIC implementation removes all software dependencies, operates at milliwatt power levels rather than the watts required by a general-purpose processor running Python, and achieves sub-millisecond latency from sensor sample to classification output. This is the architecture used in commercial cardiac monitoring devices and in the neural processing units (NPUs) found in consumer devices from Qualcomm, Apple, and Samsung.

7.2 Model Quantisation

Before any RTL implementation can begin, the trained model must be converted from 32-bit floating-point representation to a hardware-efficient fixed-point representation. Floating-point arithmetic in digital logic requires significantly more gate area than fixed-point arithmetic: a single IEEE 754 float32 multiplier occupies thousands of logic gates on a modern FPGA, making a floating-point CNN accelerator impractically large for a compact wearable chip.

Post-training INT8 quantisation via TensorFlow Lite Converter will be used to convert all 307,333 weights from float32 to signed 8-bit integers. This reduces the model weight storage from approximately 1.2 MB to 300 KB, small enough to fit entirely in on-chip SRAM without requiring external memory. The quantisation process extracts per-layer scale factors and zero-point offsets that allow the integer activations to be correctly interpreted as approximate floating-point values.

The critical verification step before writing any Verilog is to confirm that the INT8 quantised model produces identical class predictions to the float32 model on all 21,892 MIT-BIH test samples. Any discrepancy indicates quantisation error that must be investigated before proceeding. INT8 quantisation typically introduces less than 0.5% accuracy degradation on well-trained CNN classifiers, well within acceptable bounds. All intermediate computations in the RTL implementation will use Q8.8 fixed-point format (8 integer bits, 8 fractional bits, signed) with 24-bit accumulators to prevent overflow during convolution summation.

7.3 RTL Architecture

The ASIC architecture consists of five functional blocks connected in a pipeline. The SPI ADC interface receives 8-bit digital samples from an AD8232 ECG analog frontend at 125 Hz. The IIR bandpass filter

implements the fourth-order Butterworth filter (0.5 to 40 Hz) in Direct Form II transposed architecture using fixed-point arithmetic with Q2.14 coefficients computed from the Python scipy filter design. The R-peak detector uses a threshold comparator with an adaptive threshold and a minimum-distance cooldown counter that enforces the physiologically-motivated minimum RR interval constraint. The CNN accelerator implements the four convolutional blocks, global average pooling, and dense classification head as serialised multiply-accumulate units with weight storage in on-chip SRAM. The argmax output controller drives class predictions and confidence values to GPIO pins and a UART serial interface.

The RTL will be written in Verilog and targeted to the Skywater 130nm open-source process design kit (SKY130). SKY130 is a mature commercial process node made freely available by Skywater Technology and Google for academic and open-source hardware development. It includes a complete library of standard cells, SRAM macros, and IO pad cells, supported by the OpenLane RTL-to-GDSII automated physical design flow. Completed designs can be submitted to the Efabless OpenMPW shuttle program for free academic silicon fabrication, with turn-around times of approximately three to four months.

7.4 Verification Plan

Each RTL module will be developed and verified independently before integration, following established industry practice. Unit testbenches will be written in SystemVerilog for each of the following modules: SPI ADC interface, IIR biquad section, bandpass filter cascade, R-peak detector, window extractor and normaliser, MAC unit, Conv1D block (parameterised), global average pooling, dense layer, and argmax unit. Simulation will be performed using Synopsys VCS with waveform inspection in DVE, the same toolchain used in the Maven Silicon training program.

Each testbench will use Python-generated reference vectors: a known ECG input is fed to both the Python model (running at float32 and INT8 precision) and the Verilog simulation simultaneously, and every output sample is compared. This hardware-software co-verification methodology ensures bit-exact agreement between the software golden reference and the hardware implementation, providing confidence before silicon fabrication.

A full end-to-end testbench will inject the Galaxy Watch ECG sample stream as a stimulus and verify that the hardware classification output for each detected beat matches the Python model prediction. This directly links the ASIC implementation back to the real-world validation result presented in Section 6. Formal property assertions will verify that the output class is always in the valid range 0 to 4, that the done signal is never asserted during active computation, and that the window extractor never produces overlapping windows.

7.5 Implementation Roadmap

The implementation will proceed in four phases. Phase 1 (Weeks 1 to 4) covers model quantisation, weight extraction, and verification of INT8 accuracy. Phase 2 (Weeks 5 to 12) covers RTL design and unit verification of all modules, following the bottom-up order: IIR filter first (most mathematically sensitive to fixed-point precision), then R-peak detector, window extractor, MAC unit, four Conv1D blocks, global average pooling, and dense layers. Phase 3 (Weeks 13 to 16) covers top-level integration, end-to-end simulation, formal

verification, and power estimation targeting below 5 mW at 10 MHz. Phase 4 (Weeks 17 to 24) covers synthesis targeting SKY130 via OpenLane, timing closure (expected critical path through the MAC accumulation chain), physical design, design rule checking, layout versus schematic verification, and GDSII submission to the Efabless OpenMPW shuttle.

This four-phase flow mirrors the methodology used in commercial NPU and inference accelerator development at semiconductor companies. The combination of a trained machine learning model, fixed-point quantisation, Verilog RTL implementation, hardware-software co-verification, and physical design represents a complete end-to-end hardware-software co-design project.

8. Conclusion

This paper presented a 1D Convolutional Neural Network for ECG arrhythmia classification that achieves 98.44% accuracy, macro F1-score of 0.9153, and AUC-ROC of 0.9925 on the MIT-BIH Arrhythmia Database. The central finding is that class imbalance handling strategy, specifically the combination of SMOTE and focal loss is the decisive factor in achieving robust performance across all five AAMI EC57 arrhythmia classes. The same architecture without this strategy scored 29.27%, and standard class weights produced zero recall on the Fusion and Unknown classes. The combination addressed distinct mechanisms of majority-class dominance: SMOTE at the data level and focal loss at the gradient level.

A systematic investigation of Samsung Galaxy Watch 8 Classic ECG data access confirmed that Android 13 scoped storage renders raw ECG data inaccessible to external applications and ADB without device rooting. This constitutes a documented data access barrier for consumer wearable ECG research that had not been previously characterised in the machine learning literature. The only available export pathway, a formatted PDF report from Samsung Health Monitor motivated the development of the PDF digitization pipeline.

The five-stage PDF digitization pipeline successfully reconstructed ECG signals from Samsung Health Monitor PDF exports and applied the trained classifier to a personal 30-second recording. Forty-five beats were detected at 93.8 bpm, with 97.8% classified as Normal (mean confidence 84.3%), yielding a diagnosis of Normal Sinus Rhythm consistent with the Samsung clinical classification. A key technical contribution of the real-world validation was identifying and resolving the window overlap problem that arises when the RR interval is shorter than the standard MIT-BIH window half-width, which occurs at heart rates above approximately 80 bpm.

Three takeaways summarise the broader lessons of this work. First, class imbalance handling strategy matters more than model architecture: a compact 307,333-parameter model with proper training methodology outperforms more complex approaches that ignore the imbalance problem. Second, real deployment reveals challenges that benchmark evaluation cannot: the Samsung data access barrier, PDF reconstruction quality, and the high-heart-rate window overlap problem are all invisible in a standard train-test evaluation. Third, consumer health data access is a solvable engineering problem: when the expected data pathway is blocked, computer vision techniques can be applied to the available export format to recover the essential signal content.

Future work will focus on ASIC implementation of the complete pipeline in Verilog targeting the Skywater 130nm process, with the goal of deploying the system as a self-contained cardiac monitoring chip that operates continuously at milliwatt power levels.

References

- [1] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals," *Circulation*, vol. 101, no. 23, pp. e215-e220, 2000.
- [2] M. Kachuee, S. Fazeli, and M. Sarrafzadeh, "ECG Heartbeat Classification: A Deep Transferable Representation," in *Proc. IEEE ICHI*, 2018, pp. 443-444.
- [3] J. Pan and W. J. Tompkins, "A Real-Time QRS Detection Algorithm," *IEEE Trans. Biomed. Eng.*, vol. BME-32, no. 3, pp. 230-236, 1985.
- [4] P. Rajpurkar et al., "Cardiologist-Level Arrhythmia Detection and Classification in Ambulatory Electrocardiograms Using a Deep Neural Network," *Nature Medicine*, vol. 25, pp. 65-69, 2019.
- [5] A. Y. Hannun et al., "Cardiologist-Level Arrhythmia Detection and Classification in Ambulatory Electrocardiograms Using a Deep Neural Network," *Nature Medicine*, vol. 25, pp. 65-69, 2019.
- [6] H. He and E. A. Garcia, "Learning from Imbalanced Data," *IEEE Trans. Knowledge Data Eng.*, vol. 21, no. 9, pp. 1263-1284, 2009.
- [7] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321-357, 2002.
- [8] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal Loss for Dense Object Detection," in *Proc. IEEE ICCV*, 2017, pp. 2980-2988.
- [9] S. Fazeli, "Heartbeat" [Dataset], Kaggle, 2018. [Online]. Available: <https://www.kaggle.com/datasets/shayanfazeli/heartbeat>
- [10] Association for the Advancement of Medical Instrumentation, "Testing and Reporting Performance Results of Cardiac Rhythm and ST-Segment Measurement Functions," ANSI/AAMI EC57:2012.
- [11] S. Ioffe and C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," in *Proc. ICML*, 2015, pp. 448-456.
- [12] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," *J. Mach. Learn. Res.*, vol. 15, pp. 1929-1958, 2014.
- [13] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *Proc. ICLR*, 2015.
- [14] World Health Organization, "Cardiovascular Diseases (CVDs) Fact Sheet," WHO, 2023. [Online]. Available: [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))
- [15] Android Developers, "Data and file storage overview," Android Documentation, 2023. [Online]. Available: <https://developer.android.com/training/data-storage>
- [16] Skywater Technology Foundry, "SKY130 Process Design Kit," 2020. [Online]. Available: <https://github.com/google/skywater-pdk>
- [17] M. Shalan and T. Edwards, "OpenLane: The Open-Source Digital ASIC Implementation Flow," in *Proc. ICCAD*, 2020.
- [18] Efabless Corporation, "Open MPW Shuttle Program," 2021. [Online]. Available: https://efabless.com/open_shuttle_program